COMPUTER ENGINEERING

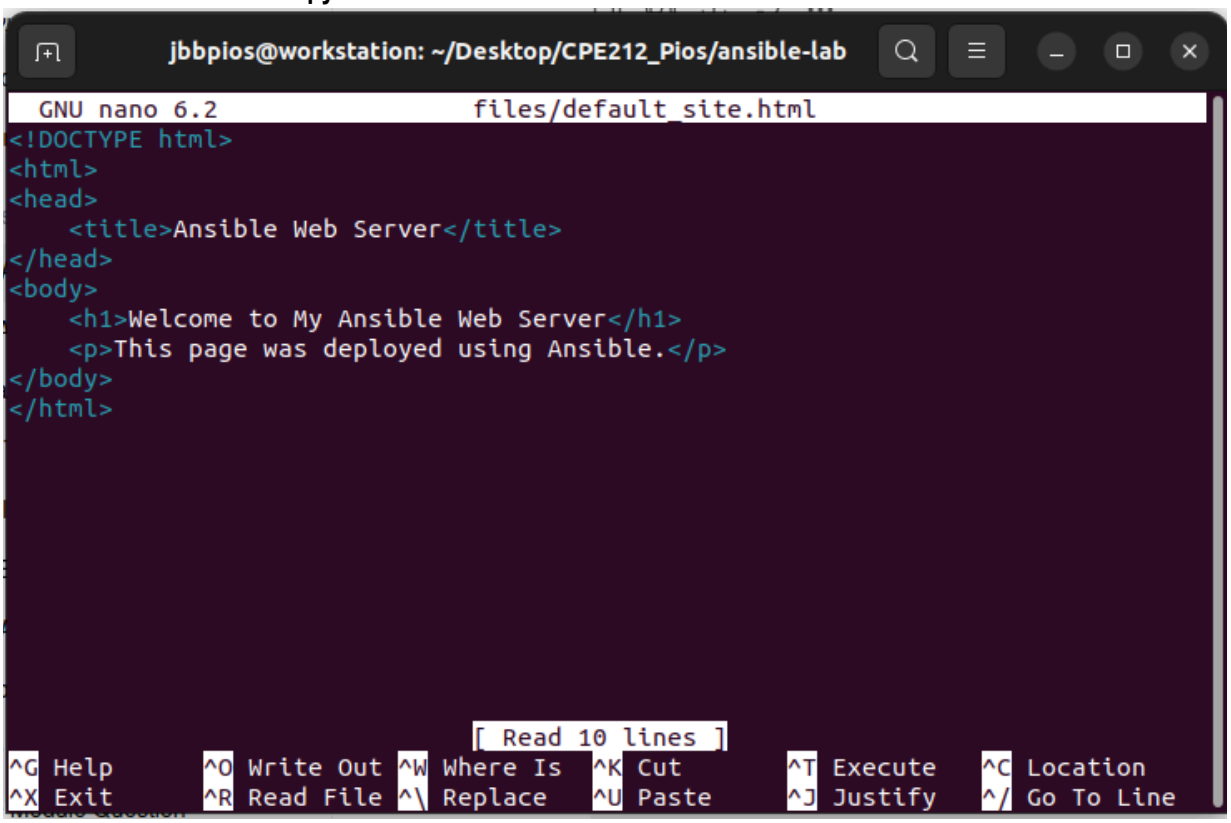
QUEZON CITY

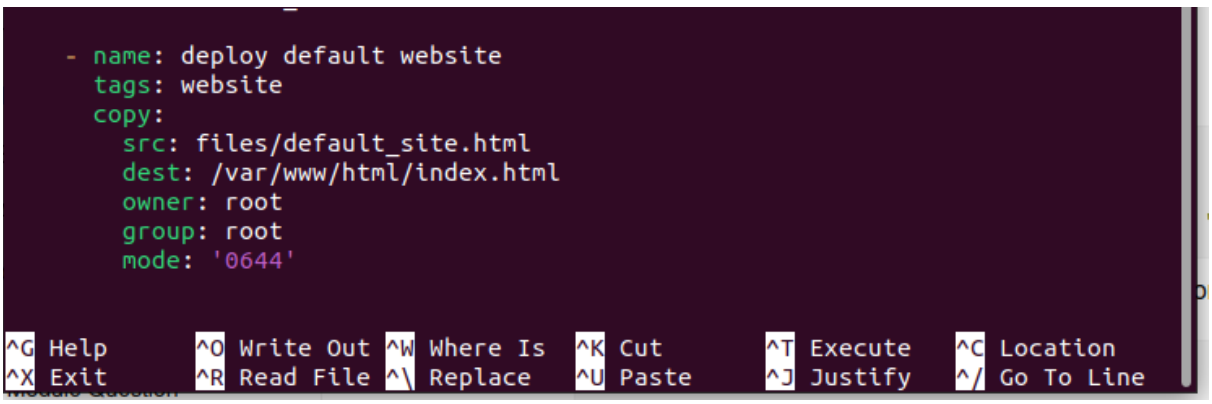
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|---|---|---|---|
| **COURSE CODE/TITLE:** | CPE 212 | **SECTION:** | CPE31S2 |
| **DATE PERFORMED:** | September 10, 2026 | **DATE SUBMITTED:** | September 10, 2026 |
| **NAME:** | Pios, Joshua Paul B. | **INSTRUCTOR:** | Engr. Jimlord M. Quejado |

**ACTIVITY NO. 7.1** **Managing Files and Creating Roles in Ansible**

## 1. PROCEDURE OUTPUT

**Task 1: Create a file and copy it to remote servers**



```
PLAY RECAP *********************************************************************
server1                    : ok=7    changed=1    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0
server2                    : ok=5    changed=2    unreachable=0    failed=0    s
kipped=2    rescued=0    ignored=0
server3                    : ok=9    changed=3    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0
```

**Task 2: Download a file and extract it to a remote server**

jbbpios@workstation: ~/Desktop/CPE212_Pios/ansible-lab

```
  GNU nano 6.2                          site.yml *
      when: ansible_distribution == "Ubuntu"

- hosts: workstations
  become: true

  tasks:

    - name: install unzip
      package:
        name: unzip
        state: present

    - name: install terraform
      unarchive:
        src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28>
        dest: /usr/local/bin
        remote_src: yes
        mode: '0755'
        owner: root
        group: root

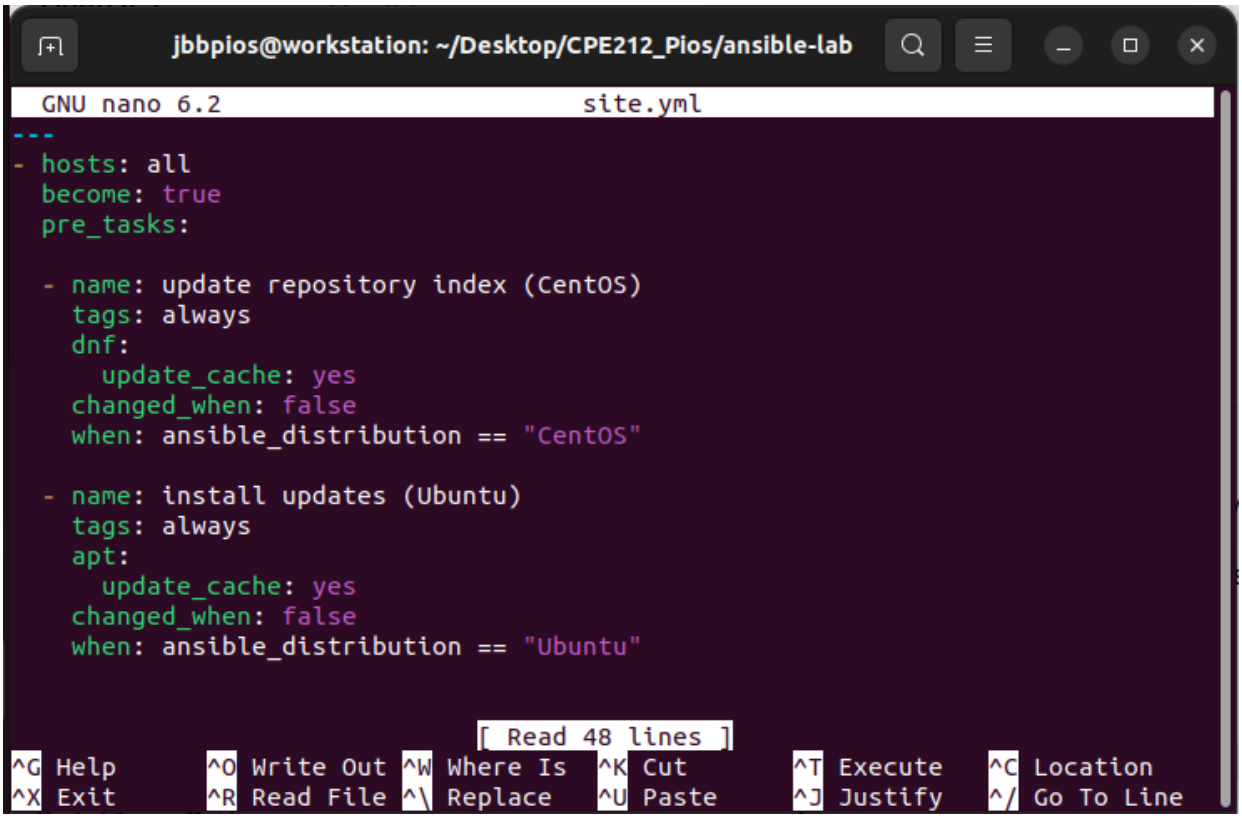
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

```
    workstations:
      hosts:
        server1:
          ansible_host: 192.168.56.101
          ansible_user: jppios
          os_type: Ubuntu
```

```
PLAY RECAP *********************************************************************
server1                    : ok=10   changed=1    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0
server2                    : ok=5    changed=1    unreachable=0    failed=0    s
kipped=2    rescued=0    ignored=0
server3                    : ok=9    changed=0    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0
```

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible server1 -i
 inventory.yaml -m command -a "terraform version"
[WARNING]: Platform linux on host server1 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
server1 | CHANGED | rc=0 >>
Terraform v0.12.28
```

**Task 3: Create roles**




```
PLAY RECAP *********************************************************************
server1                    : ok=11   changed=0    unreachable=0    failed=0    s
kipped=4    rescued=0    ignored=0
server2                    : ok=7    changed=1    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0
server3                    : ok=10   changed=0    unreachable=0    failed=0    s
kipped=4    rescued=0    ignored=0
```

The playbook successfully executed the configured Ansible roles for the base, workstation, web, database, and file servers. The roles performed the required server configurations and installed the necessary packages. The playbook completed without failed or unreachable hosts.

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

**1. What is the importance of using roles in Ansible?**
Roles organize tasks into separate and reusable sections, making playbooks easier to manage, understand, and maintain.

**2. Why is managing files important in Ansible?**
File management allows administrators to automatically copy, create, and configure files on multiple servers. This saves time and keeps server configurations consistent.

**3. What is the advantage of separating tasks into different roles?**
Separating tasks makes it easier to modify or troubleshoot a specific part of the configuration without changing the whole playbook. It also allows the same role to be reused for other servers or projects.

**Conclusion / Learnings**
In this activity, I learned how to manage files and use roles in Ansible to organize server configurations. I learned that roles make a playbook cleaner and easier to maintain because tasks for different servers can be separated into their own files. I also learned how Ansible can automatically install packages, configure web servers, manage workstations, and deploy files across multiple servers. Overall, the activity helped me understand how Ansible can simplify server management and reduce repetitive manual tasks.

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:** **N/A**

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** **N/A**

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:** **N/A**

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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